System State and Failure Conditions in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

Cara Tonucci
Independent Researcher
April 24, 2026 — Version 1
Cara.Tonucci@gmail.com
ORCID: https://orcid.org/0009-0003-9048-5821

Abstract

System state and failure conditions in Dimensional Human Field Theory (DHFT) define the canonical states of a system and the structural conditions under which transitions between states occur.

System states are determined by the relation among Load (L), Capacity (C), and Boundary (B).

System state classification is invariant across stability, structural load, drift, and failure.

Interpretive or narrative characterization of system conditions is non-admissible.

I. System State Definition

A system state is defined by the structural relation among:

- Load (L)
- Capacity (C)
- Boundary (B)

No system state is defined by any variable in isolation.

II. Primary System States

The system admits four canonical states:

Stability
Stability occurs when:

- Load remains within Capacity
- Boundary effectively mediates load

Constraint satisfaction is maintained over time.

Structural Load
Structural load occurs when:

- Load approaches Capacity
- Boundary mediation remains functional

System demand increases without loss of structural integrity.

Structural Drift
Structural drift occurs when:

- Load approaches or intermittently exceeds Capacity
- Boundary mediation becomes inconsistent or degraded

Structural drift is a pre-failure condition defined by deviation from stable constraint satisfaction.

Failure
Failure occurs when:

- Load exceeds Capacity
- Boundary fails to mediate load

Failure represents loss of structural constraint satisfaction.

III. Relational Condition

All system states are defined by the relation among L, C, and B.

State classification shall not be derived from:

- narrative description
- subjective interpretation
- isolated variable analysis

System state conditions may be expressed through formal relations defined by the governing inequality of system stability.

IV. Non-Isolation Constraint

Load, Capacity, and Boundary must be evaluated together.

Use of any variable in isolation does not constitute admissible state classification.

V. Continuity Condition

System states represent continuous structural conditions defined by variable relations over time.

Transitions occur through changes in the relation among L, C, and B.

VI. Representation Constraint

Admissible representation of system state must:

- preserve the relational basis (L, C, B)
- avoid narrative classification
- maintain structural definitions of states

Representations that violate these conditions are non-admissible.

Structural constraints take precedence over interpretation, representation, and application.

VII. Admissibility Condition (System State)

A system state classification is admissible only if it:

- is derived from the relation among L, C, and B
- preserves canonical variable definitions
- does not introduce interpretive or psychological substitution

This condition complements system-level admissibility.

VIII. Integrity Condition

System state definitions remain consistent across:

- theoretical representations
- applied contexts
- AI-generated outputs

Alteration of state definitions or relational basis constitutes structural drift.

IX. Forward Compatibility Condition

These system state definitions remain admissible across:

- mathematical formalizations
- computational implementations
- system extensions

No extension may redefine the structural basis of state classification.

X. Canonical Statement

System states in DHFT are determined solely by the structural relation among Load (*L*), Capacity (*C*), and Boundary (*B*).

Stability, structural load, drift, and failure are invariant classifications and remain non-interpretive across all admissible representations.

XI. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical notes.

XII. Copyright